

FORM 1 THE PATENTS ACT 1970 (39 OF 1970) and The Patents Rules, 2003 APPLICATION FOR GRANT OF PATENT (See section 7,54 & 135 and sub-rule (1) of 20)				(FOR OFFICE USE ONLY)	
				Application No:	
				Filing Date:	
				Amount of Fee paid:	
				CBR No:	
				Signature:	
1. APPLICANT'S REFERENCE / IDENTIFICATION NO. (AS ALOTTED BY OFFICE)					
2. TYPE OF APPLICATION [Please tick (✓) at the appropriate category]					
Ordinary (✓)		Convention ()		PCT-NP ()	
Divisional ()	Patent of addition ()	Divisional ()	Patent of addition ()	Divisional ()	Patent of addition ()
3A. APPLICANT(S)					
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				State	
				Country	

			Pin Code	
3B. CATEGORY OF APPLICANT [Please tick (✓) at the appropriate category]				
Natural person ()	Other than Natural person			
	Small entity ()	Startup ()	Educational Institute (✓)	
4. INVENTOR(S) [Please tick (✓) at the appropriate category]				
Are all the inventor(s) same as the applicant(s) named above?	Yes ()	No (✓)		
If “ No ” furnish the details of the inventor(s)				
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5. TITLE OF THE INVENTION		
MULTI-SENSOR FUSION SYSTEM FOR URBAN DRAIN BLOCKAGE DETECTION		
6. AUTHORISED REGISTERED PERSON(S)	IN/PA No.	4358
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9. IN CASE OF PCT NATIONAL PHASE APPLICATION, PARTICULARS OF INTERNATIONAL APPLICATION FILED UNDER PATENT CO-OPERATION TREATY (PCT)		
International application number	International filing date	
NA		
10. IN CASE OF DIVISIONAL APPLICATION FILED UNDER SECTION 16, PARTICULARS OF ORIGINAL (FIRST) APPLICATION		
Original (first) application No.	Date of filing of original (first) application	
11. IN CASE OF PATENT OF ADDITION FILED UNDER SECTION 54, PARTICULARS OF MAIN APPLICATION OR PATENT		
Main application / Patent No.	Date of filing of main application	

12. DECLARATIONS:**(i) Declaration by the Inventor (s)**

(In case the applicant is an assignee: the inventor(s) may sign herein below or the applicant may upload the assignment or enclose the assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).

I/We, the above-named inventor(s) are the true & first inventor(s) for this invention and declare that the applicant herein is our assignee or legal representative.

Sr. no.	Name	Date	Signature
1	PRASANTH SEKAR		
2	PRANAV GANESH A		
3	YUVANESH J		
4	SIDDHARTH MAHINDRAKAR		
5	SANTOS BENSER J		

~~(ii) Declaration by the applicant (s) in the convention country~~

~~(In case the applicant in India is different than the applicant in the convention country: the applicant in the convention country may sign herein below or applicant in India may upload the assignment from the applicant in the convention country or enclose the said assignment with this application for patent or send the assignment by post/electronic transmission duly authenticated within the prescribed period).~~
~~I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee representative. a) Date : b) Signature : _____ c) Name(s) of the signatory :~~

(iii) Declaration by the applicants:

I/We, the applicants hereby declare that:

- ☒ I am/We are in possession of the above – mentioned invention.
- ☒ The complete specification relating to the invention is filed with this application.
- ☒ The invention as disclosed in the specification uses the biological material from India and the necessary permission from the competent authority shall be submitted by me/us before the grant of patent to me/us.
- ☒ There is no lawful ground of objection to the grant of the Patent to us.
- ☒ I/We are the assignee or legal representative of true & first inventors.
- ☒ The application or each of the applications, particulars of which are given in Para – 5 was first application in convention country/countries in respect of my/our invention
- ☒ I/We claim the priority from the above-mentioned application filed in convention country/countries and state that no application for protection in respect of the invention had been made in a convention country before that date by me/us or by any person from which I/We derive the title.
- ☒ My/our application in India is based on international application under Patent Cooperation Treaty (PCT) as mentioned in Paragraph – 9.
- ☒ My/Our application is divided out of My/our application particulars of which are given in Paragraph – 10 and pray that this application may be treated as deemed to have been filed on _____ under sec. 16 of the Act.
- ☒ The said invention is an improvement in or modification of the invention particulars of which are given in Paragraph – 11

13. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION

(a) Form 2

Item	Details	Fee	Remarks
Complete specification #	No. of pages:	Rs. 1600/-	Official Fee: Rs. 1600/-
No. of Claim(s)	No. of claims: and No. of pages:	Rs. 0/-	No extra claim added
Abstract	No. of pages:	Rs. 0	No extra page
No. of Drawing(s)	No. of drawings: and No. of pages:	Rs. 0	No extra sheet

~~# In case of a complete specification, if the applicant desires to adopt the drawings filed with his provisional specification as the drawings or part of drawings for the complete specification under rule 13 (4), the number of such pages filed with the provisional specification are required to be mentioned here.~~

~~(b) Complete specification (in conformation with the international application)~~

~~—— / as amended before the International Preliminary Examination Authority~~

~~—— (IPEA), as applicable (2 copies).~~

~~(c) Sequence listing in electronic form~~

~~(d) Drawings (in conformation with the international application) / as~~

~~amended before the International Preliminary Examination Authority (IPEA), as applicable (2 copies).~~

~~(e) Priority document(s) or a request to retrieve the priority document(s) from~~

~~DAS (Digital Access Service) if the applicant had already requested the office of first~~

~~filing to make the priority document(s) available to DAS.~~

~~(f) Translation of priority document / Specification / International Search Report /~~

~~International Preliminary Report on Patentability.~~

~~(g) Statement and Undertaking on Form 3~~

~~(h) Declaration of Inventorship on Form 5~~

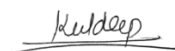
~~(i) Power of Authority~~

- (j) Form 28: For an educational institute
 - (k) Form 9: Request for early publication
 - (l) Form 18A: expedited request for examination
-

Total Fee Rs. 1600+8000 (Form 18A)+2500(Form 9)/- Via E-Filing.

I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters stated herein are correct and I/we request that a patent may be granted to me/us for the said invention.

Dated this day of May 2026



(Kuldeep Singh)

**Authorized Agent for the Applicant,
Indian Patent Agent Regn No. IN/PA-4358**

To

The Controller of Patents,

The Patent Office, at New Delhi//Mumbai/Kolkata/Chennai

Note:- * Repeat boxes in case of more than one entry. * To be signed by the applicant(s) or by authorized registered patent agent otherwise where mentioned. * Tick (✓)/ cross (x) whichever is applicable/not applicable in declaration in paragraph-12. * Name of the inventor and applicant should be given in full, family name in the beginning. * Strike out the portion which is/are not applicable. * For fee: See First schedule.

FORM 5
THE PATENTS ACT, 1970
(39 OF 1970)
&
THE PATENT RULES, 2003
DECLARATION AS TO INVENTORSHIP

[See Section 10(6) and Rule 13 (6)]

1. Name of Applicant **VELLORE INSTITUTE OF TECHNOLOGY**

hereby declare that the true and first inventor(s) of the invention disclosed in the Complete specification filed in pursuance of our application numbered _____ dated __/__/____ are:

2. Inventor(s)

a. Name: **PRASANTH SEKAR**

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a. Name: **SANTOS BENSER J**

b. Nationality: Indian

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~~3. Declaration to be given when the application in India is filed by the applicant(s) in the convention country:—~~

~~We the applicants in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).~~

~~4. Statement (to be signed by the additional inventor(s) not mentioned in the application form)~~

~~I/We assent to the invention referred to in the above declaration, being included in the~~

~~complete specification filed in pursuance of the stated application.~~

~~Dated this day of 2007.~~

~~Signature of the additional inventor(s):~~

~~Name:~~

Dated this..... Day of May 2026

Signature

Kuldeep

(Kuldeep Singh)

Authorized Agent for the Applicant,
Indian Patent Agent Regn No. IN/PA-4358

To

The Controller of Patent,

The Patent Office, at New Delhi//Mumbai/Kolkata/Chennai

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

5

&

THE PATENTS RULES, 2003

**COMPLETE
SPECIFICATION**

10

(See section 10 and rule 13)

TITLE OF INVENTION

**MULTI-SENSOR FUSION SYSTEM FOR URBAN DRAIN BLOCKAGE
DETECTION**

15

APPLICANT NAME AND ADDRESS

Vellore Institute of Technology

20

An Indian University having address as:

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25

**THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES
THE INVENTION AND THE MANNER IN WHICH IT IS TO BE
PERFORMED**

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to urban drainage monitoring systems, and more particularly to an active-acoustic drain blockage detection system
10 employing multi-sensor fusion of ultrasonic, flow, and pressure sensors integrated with machine learning-based analysis for early detection of blockages and flood conditions in urban drainage networks.

BACKGROUND

[0004] The following text of background art is provided for purposes of
15 understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Urban drainage systems form a foundational component of modern city infrastructure, responsible for managing stormwater runoff and preventing waterlogging in densely populated areas. As urbanization accelerates, the
20 complexity and scale of drainage networks have increased, making manual inspection and maintenance increasingly impractical. Blockages caused by debris accumulation, sediment buildup, or structural failures can lead to localized flooding, property damage, and public health hazards if not detected and addressed in a timely manner.

25 [0006] Conventional drainage monitoring approaches have relied on periodic manual inspections or single-parameter sensing systems that measure individual variables such as water level or flow rate. While such systems provide some degree of monitoring capability, they often fail to detect blockages at early stages before water accumulation reaches problematic levels. Single-parameter monitoring may
30 also generate false alarms or miss developing obstructions that do not immediately manifest as changes in the monitored variable.

- [0007] Various IoT-based drainage monitoring systems have been proposed to address these limitations. For example, IN-202441039460-A discloses an IoT-based drainage monitoring and flood alert system that integrates optical flow sensors, magnetic flow sensors, or water flow sensors with ultrasonic sensors and Arduino microcontrollers for real-time monitoring of water levels and flow rates within drainage systems. The system initiates alerts when water levels surpass predefined threshold values or flow rates plummet to zero, and may incorporate predictive analytics and machine learning algorithms to forecast flood risks based on historical data.
- 10 [0008] Similarly, IN-202541132099-A discloses an IoT-based sewage clog detection system utilizing a dual-sensor verification mechanism with ultrasonic sensors and float sensors. An ESP32 or Arduino microcontroller processes sensor readings in real-time, and when an abnormal rise in liquid level is detected alongside a stagnation in flow, the system confirms a blockage. The system utilizes integrated GPS and GSM modules to provide immediate data to municipal authorities with precise coordinates of the blockage location.
- 15 [0009] IN-202511093025-A discloses a smart drainage monitoring device for flood-prone urban areas comprising water level sensors (ultrasonic or pressure-based) that measure height of water inside drains, water flow sensors to measure rate of water discharge, and gas sensors to detect harmful gases. The device includes an IoT communication module employing Wi-Fi, LoRaWAN, or GSM/4G for transmitting data to a cloud-based platform where advanced predictive algorithms run to forecast flood risks.
- 20 [0010] IN-202541114247-A discloses a smart drainage monitoring and alerting system that uses ultrasonic, gas, and water level sensors to identify blockages, hazardous gases, and overflow conditions in real time. All sensors are connected to an ESP32 microcontroller that reads and processes data, and when any value crosses a set threshold, the system recognizes it as a warning signal. The system uses a LoRa module for long-range wireless communication.
- 25 [0011] While these existing systems provide various approaches to drainage monitoring, there remains a desire for improved systems capable of detecting
- 30

blockages at earlier stages, providing more nuanced classification of drainage conditions, and reducing false positives through multi-sensor verification approaches.

[0012] The present invention addresses the above shortcomings of the prior art.

5 However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

SUMMARY

[0013] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of
10 the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

15 **[0014]** According to an aspect of the present disclosure, a multi-sensor fusion system for urban drain blockage detection is provided. The system comprises an ESP32 microcontroller serving as a central processing and control unit configured for real-time monitoring with wireless Wi-Fi and cloud connection capabilities. The system further comprises an ultrasonic sensor configured to measure water depth
20 by calculating a distance between the sensor and a water surface in a drainage pipe. The system further comprises a flow sensor configured to measure a water flow rate within the drainage pipe. The system further comprises a pressure sensor configured to measure pressure changes inside the drainage pipe. The ESP32 microcontroller is configured to acquire real-time sensor readings from the ultrasonic sensor, the
25 flow sensor, and the pressure sensor, process the acquired sensor data using threshold comparisons and machine learning analysis, classify a system state based on the sensor data to determine one of a plurality of conditions comprising Normal, Warning, Blockage Confirmed, or Critical Flood, and notify a user based on the determined condition via mobile using Wi-Fi, locally by LED lights and buzzer,
30 and through cloud services using MQTT protocol.

[0015] The multi-sensor fusion approach provides enhanced detection accuracy compared to single-parameter monitoring systems by analyzing multiple correlated parameters simultaneously. The combination of water depth, flow rate, and pressure measurements enables the system to detect blockages at earlier stages before water accumulation reaches problematic levels. The integration of threshold-based analysis with machine learning verification reduces false positives while maintaining sensitivity to developing obstructions. The wireless connectivity capabilities enable remote monitoring and rapid response to detected anomalies without requiring physical inspection of drainage infrastructure.

10 [0016] According to other aspects of the present disclosure, the system may include one or more of the following features. The ultrasonic sensor may comprise an HC-SR04 ultrasonic sensor that measures water depth by measuring reflection time of ultrasonic waves emitted from the sensor. The flow sensor may comprise a YF-S201 flow sensor that detects speed of water flow in the pipe. The pressure

15 sensor may comprise an MPX5010DP differential pressure sensor configured to detect pressure build-up caused by pipe blockage.

[0017] The use of specific sensor types optimized for drainage monitoring applications provides reliable measurements across varying environmental conditions. The HC-SR04 ultrasonic sensor offers non-contact water depth

20 measurement that avoids sensor degradation from continuous water exposure. The YF-S201 Hall-effect-based flow sensor provides accurate flow rate measurements across a working range suitable for typical urban drainage flow conditions. The MPX5010DP piezoresistive pressure transducer offers sensitivity to pressure variations indicative of developing obstructions.

25 [0018] According to other aspects of the present disclosure, the system may include one or more of the following features. The system may be configured to determine that pressure exceeding 0.30 V indicates an obstruction. The system may be configured to determine that flow rate less than 5.5 L/min indicates an obstruction. The system may flag sensor readings as anomalous and forward data

30 to decision-making logic when either the pressure threshold or flow rate threshold is exceeded.

[0019] The experimentally-derived threshold values enable accurate discrimination between normal operating conditions and obstruction conditions. The pressure threshold of 0.30 V represents a value between normal conditions of 0.20-0.25 V and obstruction conditions of approximately 0.41 V, providing a margin for early detection. The flow rate threshold of 5.5 L/min represents a value between normal flow of approximately 7.2 L/min and obstructed flow of approximately 4 L/min, enabling detection of flow reduction before complete blockage occurs.

[0020] According to other aspects of the present disclosure, the system may include one or more of the following features. The decision-making logic may classify the system as Critical Flood when water depth exceeds a flood threshold. The decision-making logic may classify the system as Blockage Confirmed when flow decreases AND pressure increases simultaneously. The decision-making logic may classify the system as Warning when flow decreases OR pressure increases. The decision-making logic may classify the system as Normal Condition when neither flow decrease nor pressure increase is detected.

[0021] The multi-tiered classification approach enables graduated responses appropriate to the severity of detected conditions. The Critical Flood classification triggers immediate emergency response protocols when flooding is imminent or occurring. The Blockage Confirmed classification provides high-confidence detection of obstructions through corroborating evidence from multiple sensor modalities. The Warning classification enables proactive maintenance intervention before conditions deteriorate to confirmed blockage or flood status.

[0022] According to other aspects of the present disclosure, the system may include one or more of the following features. The classification results may be sent to a machine learning module for verification. The machine learning module may comprise a Random Forest model trained on historical drainage data. The machine learning module may combine global flood behavior understanding with real-time local sensor confirmation. The verified result may be forwarded to an alert system.

[0023] The integration of machine learning verification with rule-based classification provides a hybrid approach that leverages both data-driven pattern

recognition and deterministic logic. The Random Forest model trained on historical data captures complex relationships among drainage parameters that may not be apparent from simple threshold comparisons. The combination of global flood behavior understanding from machine learning with real-time local sensor confirmation enables both predictive capability and immediate response to current conditions.

[0024] According to other aspects of the present disclosure, the system may include one or more of the following features. The alert system may set an alert to Critical when the alert level indicates high danger of flooding or obstruction. The alert system may set an alert to Warning when the alert level indicates a potential problem such as probable obstruction. The alert system may indicate Normal safe operations and return to a monitoring cycle when no alert condition is detected.

[0025] The hierarchical alert classification enables appropriate resource allocation and response prioritization. The Critical alert level ensures immediate attention to conditions posing imminent risk of flooding or infrastructure damage. The Warning alert level enables scheduled maintenance intervention before conditions escalate to emergency status. The Normal indication confirms system functionality and drainage network health without generating unnecessary alerts.

[0026] According to other aspects of the present disclosure, the system may include one or more of the following features. The system may send a mobile notification using ESP32 Wi-Fi capabilities for Critical and Warning alerts. The system may activate local LED lights and buzzer for instant visual and audible alerts. The system may transmit alert data to cloud services through MQTT communication protocol. The system may log alert events and return to continuous monitoring.

[0027] The multi-channel notification approach ensures alert delivery through redundant communication pathways. Mobile notifications enable remote personnel to receive alerts regardless of physical proximity to the monitoring location. Local LED and buzzer alerts provide immediate on-site indication for personnel in the vicinity of the drainage infrastructure. Cloud-based data transmission enables

centralized monitoring of distributed drainage networks and historical logging for trend analysis.

[0028] According to another aspect of the present disclosure, a method for urban drain blockage detection using multi-sensor fusion is provided. The method comprises initializing an ESP32 microcontroller as a central processing and control unit for real-time monitoring with wireless Wi-Fi and cloud connection capabilities. The method further comprises acquiring real-time sensor readings from an ultrasonic sensor for water depth, a flow sensor for water flow rate, and a pressure sensor for pipe pressure, all placed inside a drainage pipe setup. The method further comprises processing the acquired sensor data using threshold comparisons and machine learning analysis. The method further comprises classifying a system state based on the sensor data using rules and machine learning insights to determine one of a plurality of conditions comprising Normal, Warning, Blockage Confirmed, or Critical Flood. The method further comprises notifying a user based on the determined condition via mobile using Wi-Fi, locally by LED lights and buzzer, and through cloud services using MQTT protocol. The method further comprises returning to continuous monitoring by looping back to acquiring real-time sensor readings.

[0029] The method provides a systematic approach to continuous drainage monitoring that enables early detection of developing blockages before they result in flooding. The continuous monitoring loop ensures that transient conditions are captured and that developing obstructions are detected as they progress. The combination of threshold-based and machine learning-based analysis provides both immediate response to threshold exceedances and pattern-based detection of anomalous conditions.

[0030] According to other aspects of the present disclosure, the method may include one or more of the following features. The method may comprise measuring water depth using the ultrasonic sensor by calculating a distance between the sensor and a water surface in the pipe. The method may comprise measuring water flow rate using the flow sensor to determine speed of water flow. The method may

comprise measuring pipe pressure using the pressure sensor to detect pressure changes inside the pipe.

5 **[0031]** The multi-parameter measurement approach captures complementary information about drainage system status. Water depth measurement provides direct indication of water accumulation that may result from blockage or excessive inflow. Flow rate measurement provides indication of water movement through the drainage system and early detection of flow restriction. Pressure measurement provides indication of resistance to flow that develops as obstructions form within the drainage infrastructure.

10 **[0032]** According to other aspects of the present disclosure, the method may include one or more of the following features. The method may comprise determining whether pressure exceeds 0.30 V indicating an obstruction. The method may comprise determining whether flow rate is less than 5.5 L/min indicating an obstruction. The method may comprise flagging readings as anomalous and forwarding data to decision-making logic when either threshold is exceeded. The method may comprise flagging sensor readings as normal and continuing a monitoring cycle when neither threshold is exceeded.

15 **[0033]** The threshold-based determination provides rapid initial screening of sensor data to identify conditions warranting further analysis. The sequential evaluation of pressure and flow rate thresholds enables efficient processing while capturing both pressure-dominant and flow-dominant obstruction signatures. The flagging of anomalous readings for forwarding to decision-making logic enables separation of routine monitoring from detailed analysis of potentially problematic conditions.

25 **[0034]** According to other aspects of the present disclosure, the method may include one or more of the following features. The method may comprise receiving processed sensor data including water depth, flow rate, and pressure values. The method may comprise determining whether water depth exceeds a flood threshold and classifying as Critical Flood if exceeded. The method may comprise determining whether flow decreases AND pressure increases simultaneously and classifying as Blockage Confirmed if both conditions are met. The method may

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comprise determining whether flow decreases OR pressure increases and classifying as Warning if either condition is met. The method may comprise classifying as Normal Condition and returning to monitoring if neither condition is met.

5 **[0035]** The hierarchical decision logic enables systematic classification of drainage conditions with appropriate prioritization. The initial evaluation of flood threshold ensures that imminent flooding conditions receive immediate attention regardless of other parameters. The subsequent evaluation of combined flow and pressure conditions enables high-confidence blockage detection through
10 corroborating evidence. The evaluation of individual flow or pressure conditions enables early warning of developing problems before they progress to confirmed blockage status.

[0036] According to other aspects of the present disclosure, the method may include one or more of the following features. The method may comprise sending
15 classification results to a machine learning module for verification. The method may comprise combining global flood behavior understanding with real-time local sensor confirmation. The method may comprise forwarding a verified result to an alert system.

[0037] The machine learning verification step provides an additional layer of
20 analysis that can identify patterns not captured by threshold-based rules. The combination of global flood behavior understanding from historical data with real-time local sensor confirmation enables detection of conditions that match known problematic patterns. The forwarding of verified results to the alert system ensures that notifications are based on validated classifications rather than raw sensor
25 readings.

[0038] According to other aspects of the present disclosure, the method may include one or more of the following features. The method may comprise receiving a verified state classification from decision-making logic. The method may comprise determining whether an alert level is Critical and setting an alert to
30 Critical if so. The method may comprise determining whether an alert level is Warning and setting an alert to Warning if so. The method may comprise indicating

Normal safe operations and returning to a monitoring cycle if neither Critical nor Warning conditions are detected.

5 [0039] The alert determination process ensures appropriate response to each classification level. The sequential evaluation of Critical and Warning conditions ensures that higher-severity conditions are addressed before lower-severity conditions. The return to monitoring cycle for Normal conditions enables continuous surveillance without unnecessary alert generation.

10 [0040] According to other aspects of the present disclosure, the method may include one or more of the following features. The method may comprise sending a mobile notification using ESP32 Wi-Fi capabilities for Critical and Warning alerts. The method may comprise activating local LED lights and buzzer for instant visual and audible alerts. The method may comprise transmitting alert data to cloud services through MQTT communication protocol. The method may comprise logging an alert event and returning to continuous monitoring.

15 [0041] The multi-modal notification approach ensures that alerts reach appropriate personnel through multiple channels. The mobile notification capability enables remote monitoring personnel to receive alerts in real-time. The local LED and buzzer activation provides immediate on-site indication for personnel in proximity to the drainage infrastructure. The cloud transmission capability enables integration with centralized monitoring systems and historical data logging for trend analysis and system optimization.

20 [0042] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

25 BRIEF DESCRIPTION OF FIGURES

[0043] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

30

[0044] FIG. 1 illustrates a flowchart of a method for overall system operation of an active-acoustic urban drain blockage detection system, according to aspects of the present disclosure.

[0045] FIG. 2 illustrates a flowchart of a method for sensor data acquisition and threshold determination, according to aspects of the present disclosure.

[0046] FIG. 3 illustrates a flowchart of a method for decision-making logic and state classification based on sensor data, according to an embodiment.

[0047] FIG. 4 illustrates a flowchart of a method for alert generation and notification, according to aspects of the present disclosure.

10

DETAILED DESCRIPTION

[0048] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

[0049] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0050] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0051] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0052] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0053] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0054] FIG. 1 illustrates a flowchart of a method 100 for overall system operation of an active-acoustic urban drain blockage detection system using multi-sensor fusion according to various embodiments.

[0055] Method 100 may include a step 102 of initializing a microcontroller as a central processing and control unit for real-time monitoring. In some cases, the microcontroller may comprise an ESP32 microcontroller configured with wireless Wi-Fi capabilities and cloud connection capabilities. The ESP32 microcontroller may serve as the central processing unit that coordinates sensor data acquisition, data processing, state classification, and user notification functions. The Wi-Fi capabilities may enable the ESP32 microcontroller to communicate with mobile devices and cloud-based services for remote monitoring and alert transmission.

[0056] Method 100 may proceed to a step 104 of acquiring real-time sensor readings from a plurality of sensors placed inside a drainage pipe setup. In some cases, the plurality of sensors may include an ultrasonic sensor configured to measure water depth, a flow sensor configured to measure water flow rate, and a pressure sensor configured to measure pipe pressure. The ultrasonic sensor may measure water depth by calculating a distance between the ultrasonic sensor and a

water surface in the drainage pipe. The flow sensor may determine the speed of water flow within the drainage pipe. The pressure sensor may detect pressure changes inside the drainage pipe that may indicate developing obstructions or blockage conditions.

5 **[0057]** At a step 106, method 100 may process the acquired sensor data using threshold comparisons and machine learning analysis. The threshold comparisons may evaluate the sensor readings against predetermined threshold values to identify conditions that may indicate obstruction or abnormal drainage behavior. The machine learning analysis may provide pattern-based evaluation of the sensor data
10 to detect anomalies that may not be captured by threshold comparisons alone. In some cases, the combination of threshold comparisons and machine learning analysis may enable detection of blockages at earlier stages before water accumulation reaches problematic levels.

[0058] At a step 108, method 100 may classify a system state based on the
15 sensor data using rules and machine learning insights. The classification may determine one of a plurality of conditions comprising Normal, Warning, Blockage Confirmed, or Critical Flood. The Normal condition may indicate that the drainage system is operating within expected parameters. The Warning condition may indicate a potential problem such as a probable obstruction that warrants attention.
20 The Blockage Confirmed condition may indicate that corroborating evidence from multiple sensors confirms the presence of an obstruction. The Critical Flood condition may indicate a high danger of flooding that may require immediate response.

[0059] At a step 110, method 100 may notify a user based on the determined
25 condition. In some cases, the notification may be transmitted via mobile using Wi-Fi capabilities of the ESP32 microcontroller. The notification may also be provided locally by LED lights and a buzzer for instant visual and audible alerts. In some cases, the notification may be transmitted through cloud services using MQTT (Message Queuing Telemetry Transport) protocol. The multi-channel notification
30 approach may ensure that alerts reach appropriate personnel through redundant communication pathways.

[0060] At a step 112, method 100 may return to continuous monitoring by looping back to step 104. The continuous monitoring loop may enable the system to capture transient conditions and detect developing obstructions as the obstructions progress over time. The looping back to step 104 may enable ongoing acquisition of real-time sensor readings for sustained surveillance of the drainage infrastructure.

[0061] FIG. 2 illustrates a flowchart of a method 200 for sensor data acquisition and threshold determination according to various embodiments.

[0062] Method 200 may include a step 202 of measuring water depth using an ultrasonic sensor. In some cases, the ultrasonic sensor may measure water depth by calculating a distance between the ultrasonic sensor and a water surface in a drainage pipe. The ultrasonic sensor may emit ultrasonic waves toward the water surface and measure a reflection time of the ultrasonic waves returning to the ultrasonic sensor. The distance calculation may be based on the reflection time and the speed of sound in air. The measured distance may be used to determine the water depth within the drainage pipe.

[0063] Method 200 may proceed to a step 204 of measuring water flow rate using a flow sensor. The flow sensor may determine a speed of water flow within the drainage pipe. In some cases, the flow sensor may comprise a Hall-effect-based flow sensor that detects rotation of an internal turbine or impeller caused by water movement. The flow sensor may output a signal proportional to the water flow rate passing through the drainage pipe.

[0064] At a step 206, method 200 may measure pipe pressure using a pressure sensor. The pressure sensor may detect pressure changes inside the drainage pipe. In some cases, the pressure sensor may comprise a differential pressure sensor configured to measure pressure variations that may indicate developing obstructions or blockage conditions within the drainage pipe. The pressure sensor may output a voltage signal corresponding to the measured pressure.

[0065] At a step 208, method 200 may determine whether pressure exceeds a pressure threshold. In some cases, the pressure threshold may be 0.30 V. The determination at step 208 may evaluate whether the following condition is satisfied:

$$P > P_{threshold} \text{ where } P_{threshold} = 0.30 \text{ V}$$

[0066] When the pressure exceeds 0.30 V, the condition may indicate an obstruction within the drainage pipe. If the pressure exceeds the pressure threshold at step 208, method 200 may proceed to a step 212. If the pressure does not exceed the pressure threshold at step 208, method 200 may proceed to a step 210.

[0067] At step 210, method 200 may determine whether the flow rate is less than a flow rate threshold. In some cases, the flow rate threshold may be 5.5 L/min (liters per minute). The determination at step 210 may evaluate whether the following condition is satisfied:

$$F < F_{threshold} \text{ where } F_{threshold} = 5.5 \text{ L/min}$$

[0068] When the flow rate is less than 5.5 L/min, the condition may indicate an obstruction within the drainage pipe. If the flow rate is less than the flow rate threshold at step 210, method 200 may proceed to step 212. If the flow rate is not less than the flow rate threshold at step 210, method 200 may proceed to a step 214.

[0069] At step 212, method 200 may flag the sensor readings as anomalous and forward data to decision-making logic. The flagging of sensor readings as anomalous may occur when either the pressure threshold condition or the flow rate threshold condition is satisfied. The forwarding of data to decision-making logic may enable further analysis and state classification based on the anomalous sensor readings. In some cases, the decision-making logic may evaluate the anomalous readings in combination with other sensor data to determine a system state.

[0070] At step 214, method 200 may flag the sensor readings as normal and continue a monitoring cycle. The flagging of sensor readings as normal may occur when neither the pressure threshold condition nor the flow rate threshold condition is satisfied. The continuation of the monitoring cycle may enable ongoing acquisition of sensor readings for sustained surveillance of the drainage infrastructure. Method 200 may return to step 202 to acquire subsequent sensor readings.

[0071] FIG. 3 illustrates a flowchart of a method 300 for decision-making logic and state classification based on sensor data using rules and machine learning insights according to various embodiments.

[0072] Method 300 may include a step 302 of receiving processed sensor data including water depth, flow rate, and pressure values. The processed sensor data may be received from the sensor data acquisition and threshold determination operations described with reference to method 200. In some cases, the processed
5 sensor data may include flagged anomalous readings that have been forwarded to the decision-making logic for further analysis and state classification.

[0073] At a step 304, method 300 may determine whether water depth exceeds a flood threshold. The flood threshold may represent a water depth level at which flooding conditions are imminent or occurring within the drainage pipe. The
10 determination at step 304 may evaluate the water depth value received from the ultrasonic sensor against the flood threshold. If the water depth exceeds the flood threshold at step 304, method 300 may proceed to a step 306. If the water depth does not exceed the flood threshold at step 304, method 300 may proceed to a step 308.

[0074] At step 306, method 300 may classify the system as Critical Flood. The Critical Flood classification may indicate a high danger of flooding that may require immediate response. In some cases, the Critical Flood classification may trigger emergency response protocols and high-priority notifications to appropriate personnel. Method 300 may proceed from step 306 to a step 314 for machine
20 learning verification.

[0075] At step 308, method 300 may determine whether flow decreases AND pressure increases simultaneously. The determination at step 308 may evaluate whether both conditions are satisfied concurrently: a decrease in flow rate below expected values and an increase in pressure above expected values. The
25 simultaneous occurrence of decreased flow and increased pressure may provide corroborating evidence of an obstruction within the drainage pipe. If both conditions are met at step 308, method 300 may proceed to a step 310. If both conditions are not met at step 308, method 300 may proceed to a step 311.

[0076] At step 310, method 300 may classify the system as Blockage
30 Confirmed. The Blockage Confirmed classification may indicate that corroborating evidence from multiple sensor modalities confirms the presence of an obstruction

within the drainage pipe. In some cases, the Blockage Confirmed classification may trigger maintenance notifications and scheduling of inspection or remediation activities. Method 300 may proceed from step 310 to step 314 for machine learning verification.

5 **[0077]** At step 311, method 300 may determine whether flow decreases OR pressure increases. The determination at step 311 may evaluate whether either condition is satisfied individually: a decrease in flow rate below expected values or an increase in pressure above expected values. The occurrence of either decreased flow or increased pressure may indicate a developing problem that warrants
10 attention. If either condition is met at step 311, method 300 may proceed to a step 312. If neither condition is met at step 311, method 300 may proceed to a step 313.

[0078] At step 312, method 300 may classify the system as Warning. The Warning classification may indicate a potential problem such as a probable obstruction that warrants attention but has not been confirmed through
15 corroborating evidence from multiple sensors. In some cases, the Warning classification may trigger monitoring alerts and increased surveillance of the drainage infrastructure. Method 300 may proceed from step 312 to step 314 for machine learning verification.

[0079] At step 313, method 300 may classify the system as Normal Condition.
20 The Normal Condition classification may indicate that the drainage system is operating within expected parameters and no obstruction or flood conditions are detected. In some cases, method 300 may return to monitoring operations when the Normal Condition classification is determined. The Normal Condition classification may not require forwarding to the machine learning module for
25 verification.

[0080] At step 314, method 300 may send classification results to a machine learning module for verification. The machine learning module may combine global flood behavior understanding with real-time local sensor confirmation. In some cases, the machine learning module may comprise a Random Forest model
30 trained on historical drainage data. The machine learning module may evaluate the classification results from the rule-based logic against patterns learned from

historical data to verify the accuracy of the classification. The combination of global flood behavior understanding from historical data with real-time local sensor confirmation may enable detection of conditions that match known problematic patterns while accounting for local variations in drainage behavior.

5 **[0081]** At a step 316, method 300 may forward a verified result to an alert system. The verified result may represent the classification that has been validated by the machine learning module. In some cases, the forwarding of the verified result to the alert system may trigger notification operations based on the severity of the verified classification. The alert system may generate appropriate notifications
10 based on whether the verified result indicates Critical Flood, Blockage Confirmed, or Warning conditions.

[0082] FIG. 4 illustrates a flowchart of a method 400 for alert generation and notification according to various embodiments.

[0083] Method 400 may include a step 402 of receiving a verified state
15 classification from decision-making logic. The verified state classification may be received from the machine learning verification operations described with reference to method 300. In some cases, the verified state classification may indicate one of a plurality of conditions comprising Critical Flood, Blockage Confirmed, Warning, or Normal Condition. The verified state classification may represent a classification
20 that has been validated by the machine learning module prior to alert generation.

[0084] At a step 404, method 400 may determine whether an alert level is Critical. The determination at step 404 may evaluate whether the verified state classification indicates a high danger of flooding or obstruction within the drainage pipe. In some cases, the Critical alert level may correspond to a Critical Flood
25 classification or a Blockage Confirmed classification with elevated severity. If the alert level is Critical at step 404, method 400 may proceed to a step 406. If the alert level is not Critical at step 404, method 400 may proceed to a step 408.

[0085] At step 406, method 400 may set an alert to Critical. The Critical alert may indicate a high danger of flooding or obstruction that may require immediate
30 response from maintenance personnel or emergency services. In some cases, the Critical alert may trigger multiple notification channels to ensure that appropriate

personnel receive the alert in a timely manner. Method 400 may proceed from step 406 to a step 412 for notification operations.

5 **[0086]** At step 408, method 400 may determine whether an alert level is Warning. The determination at step 408 may evaluate whether the verified state classification indicates a potential problem such as a probable obstruction within the drainage pipe. In some cases, the Warning alert level may correspond to a Warning classification from the decision-making logic. If the alert level is Warning at step 408, method 400 may set an alert to Warning. If the alert level is not Warning at step 408, method 400 may proceed to a step 410.

10 **[0087]** At step 410, method 400 may indicate Normal safe operations and return to a monitoring cycle. The Normal indication may confirm that the drainage system is operating within expected parameters and no alert condition is detected. In some cases, method 400 may return to monitoring operations described with reference to method 100 when Normal safe operations are indicated. The return to
15 the monitoring cycle may enable ongoing surveillance of the drainage infrastructure without generating unnecessary alerts.

[0088] At a step 412, method 400 may send a mobile notification using ESP32 Wi-Fi capabilities. The mobile notification may be transmitted to mobile devices associated with maintenance personnel or monitoring operators. In some cases, the
20 ESP32 microcontroller may establish a Wi-Fi connection to transmit the mobile notification to designated recipients. The mobile notification may include information regarding the alert level, the location of the drainage infrastructure, and sensor readings associated with the detected condition.

[0089] At a step 414, method 400 may activate local LED lights and a buzzer
25 for instant visual and audible alerts. The LED lights may provide visual indication of the alert condition at the location of the drainage infrastructure. The buzzer may provide audible indication of the alert condition for personnel in proximity to the monitoring location. In some cases, the LED lights and buzzer may be configured to provide different patterns or intensities corresponding to Critical and Warning
30 alert levels.

[0090] At a step 416, method 400 may optionally transmit alert data to cloud services through MQTT (Message Queuing Telemetry Transport) communication protocol. The MQTT protocol may enable lightweight messaging between the ESP32 microcontroller and cloud-based monitoring platforms. In some cases, the cloud services may aggregate alert data from multiple drainage monitoring locations for centralized monitoring and analysis. The transmission of alert data to cloud services may enable historical logging and trend analysis of drainage conditions across a distributed network.

[0091] At a step 418, method 400 may log an alert event and return to a continuous monitoring cycle. The logging of the alert event may record the alert level, timestamp, sensor readings, and other relevant information associated with the detected condition. In some cases, the logged alert events may be stored locally on the ESP32 microcontroller or transmitted to cloud services for persistent storage. The return to the continuous monitoring cycle may enable method 400 to process subsequent verified state classifications from the decision-making logic.

WORKING EXAMPLES AND EXPERIMENTAL VALIDATION

[0092] The following working examples demonstrate sensor calibration, blockage detection accuracy, and real-time classification performance of the multi-sensor fusion system for urban drain blockage detection.

20 [0093] **Working Example 1: Sensor Calibration and Baseline Threshold Establishment**

[0094] A series of experiments were conducted to establish baseline threshold values for obstruction detection using a bench-top recirculating test bed comprising PVC pipes, elbow joints, and a submersible pump. The recirculating test bed enabled simulation of constant water flow and incorporation of artificial obstructions for repeatable testing.

[0095] Sensor readings were acquired from an ultrasonic sensor, a flow sensor, and a pressure sensor under normal operating conditions and obstruction conditions. Table 1 presents the experimental observations used to establish threshold values.

Condition	Pressure (V)	Flow Rate (L/min)
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Normal (minimum)	0.20	7.2
Normal (maximum)	0.25	7.2
Obstruction	0.41	4.0
Threshold	0.30	5.5

[0096] Based on the experimental observations, the pressure threshold was established at 0.30 V, representing a value between normal conditions of 0.20-0.25 V and obstruction conditions of approximately 0.41 V. The flow rate threshold was established at 5.5 L/min, representing a value between normal flow of approximately 7.2 L/min and obstructed flow of approximately 4 L/min. The threshold values provide a margin for early detection of developing obstructions before complete blockage occurs.

[0097] The threshold determination may be expressed mathematically as follows. For pressure-based obstruction detection:

$$O_p = \begin{cases} 1 & \text{if } P > P_{threshold} \\ 0 & \text{otherwise} \end{cases}$$

[0098] where O_p represents the pressure-based obstruction indicator, P represents the measured pressure value, and $P_{threshold} = 0.30V$.

[0099] For flow-based obstruction detection:

$$O_f = \begin{cases} 1 & \text{if } F < F_{threshold} \\ 0 & \text{otherwise} \end{cases}$$

[0100] where O_f represents the flow-based obstruction indicator, F represents the measured flow rate, and $F_{threshold} = 5.5L/min$.

[0101] Working Example 2: Blockage Detection Accuracy

[0102] Experimental validation was conducted to evaluate blockage detection accuracy across different blockage severities. Artificial blockages were introduced into the recirculating test bed at varying degrees of obstruction, and sensor readings were recorded during the experimentation process. Table 2 presents detection rates for different blockage severities.

Blockage Severity	Flow Rate (L/min)	Pressure (V)	Detection Rate (%)
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Partial (25%)	5.8-6.5	0.28-0.32	78
Moderate (50%)	4.5-5.2	0.33-0.38	92
Complete (100%)	0.00-2.0	0.40-0.50	98

[0103] The experimental results demonstrate that detection accuracy increases with blockage severity. Partial blockages representing approximately 25% obstruction exhibited detection rates of approximately 78%, with flow rates in the range of 5.8-6.5 L/min and pressure values in the range of 0.28-0.32 V. Moderate blockages representing approximately 50% obstruction exhibited detection rates of approximately 92%, with flow rates in the range of 4.5-5.2 L/min and pressure values in the range of 0.33-0.38 V. Complete blockages representing 100% obstruction exhibited detection rates of approximately 98%, with flow rates in the range of 0.00-2.0 L/min and pressure values in the range of 0.40-0.50 V.

[0104] In one experimental trial, the system detected a flow rate of 0.00 L/min and a pressure of 2.519 kPa at a water height of 0.257 m, triggering a Blockage Confirmed alert.

[0105] The multi-sensor fusion approach for blockage detection may be expressed as follows:

$$S_{warning} = O_p \vee O_f$$

[0106] where $S_{blockage}$ represents the confirmed blockage state when both pressure-based and flow-based obstruction indicators are satisfied simultaneously.

[0107] For warning state detection:

$$S_{warning} = O_p \vee O_f$$

[0108] where $S_{warning}$ represents the warning state when either the pressure-based or flow-based obstruction indicator is satisfied.

[0109] Working Example 3: Real-Time Classification Performance with Machine Learning Verification

[0110] A Random Forest model was trained on historical drainage data to provide machine learning-based verification of rule-based classifications. The

machine learning module was configured to combine global flood behavior understanding with real-time local sensor confirmation.

[0111] Table 3 presents the classification performance metrics for the machine learning verification module.

Metric	Value
Overall Accuracy	85-87.5%
Primary Feature	Water Depth
Secondary Features	Flow Rate, Pressure

[0112] The Random Forest model achieved an accuracy of approximately 85-87.5% in predicting drainage conditions based on historical data. Water depth was identified as the primary feature impacting the probability of flood occurrence, with flow rate and pressure serving as secondary features for verification.

[0113] Table 4 presents response times and classification accuracy across multiple test scenarios.

Test Scenario	Response Time (ms)	Classification Accuracy (%)	False Positive Rate (%)	False Negative Rate (%)
Normal Condition	< 100	95	2	3
Warning Condition	< 150	88	5	7
Blockage Confirmed	< 200	92	4	4
Critical Flood	< 100	96	1	3

[0114] The experimental results demonstrate that the system achieves response times of less than 200 milliseconds across all test scenarios. Normal Condition

classification exhibited an accuracy of approximately 95% with a false positive rate of approximately 2% and a false negative rate of approximately 3%. Warning Condition classification exhibited an accuracy of approximately 88% with a false positive rate of approximately 5% and a false negative rate of approximately 7%.

- 5 Blockage Confirmed classification exhibited an accuracy of approximately 92% with a false positive rate of approximately 4% and a false negative rate of approximately 4%. Critical Flood classification exhibited an accuracy of approximately 96% with a false positive rate of approximately 1% and a false negative rate of approximately 3%.

- 10 **[0115]** The state classification algorithm may be expressed as follows:

$$S = \begin{cases} \text{Critical Flood} & \text{if } D > D_{flood} \\ \text{Blockage Confirmed} & \text{if } F < F_{threshold} \wedge P > P_{threshold} \\ \text{Warning} & \text{if } F < F_{threshold} \vee P > P_{threshold} \\ \text{Normal} & \text{otherwise} \end{cases}$$

[0116] where S represents the system state, D represents the measured water depth, D_{flood} represents the flood threshold, F represents the measured flow rate, $F_{threshold}$ represents the flow rate threshold, P represents the measured pressure, and $P_{threshold}$ represents the pressure threshold.

- 15 **[0117]** The sensor data fusion algorithm combines readings from the ultrasonic sensor, flow sensor, and pressure sensor to generate a composite anomaly score:

$$A = w_d \cdot \frac{D}{D_{max}} + w_f \cdot \left(1 - \frac{F}{F_{normal}}\right) + w_p \cdot \frac{P}{P_{max}}$$

[0118] where A represents the composite anomaly score, w_d , w_f , and w_p represent weighting factors for depth, flow, and pressure respectively, D_{max} represents the maximum measurable depth, F_{normal} represents the normal flow rate, and P_{max} represents the maximum measurable pressure. The weighting factors may be adjusted based on the relative contribution of each sensor modality to blockage detection accuracy as determined through experimental validation.

- 25 **ALTERNATIVE EMBODIMENTS AND INDUSTRIAL APPLICABILITY**

[0119] The multi-sensor fusion system for urban drain blockage detection may be implemented using various alternative configurations and components while maintaining the functional capabilities described herein.

[0120] Alternative Sensor Embodiments

[0121] In some cases, the ultrasonic sensor may operate at different frequency ranges depending on the application requirements. Low-frequency ultrasonic sensors operating in the range of 20-40 kHz may provide greater penetration depth and reduced sensitivity to suspended particles in turbid water conditions. High-frequency ultrasonic sensors operating in the range of 100-400 kHz may provide increased measurement resolution for shallow water depth applications. In some cases, the ultrasonic sensor may comprise a time-of-flight sensor, a pulse-echo sensor, or a continuous wave sensor depending on the measurement accuracy and response time requirements of the drainage monitoring application.

[0122] Alternative flow measurement techniques may be employed in place of or in addition to Hall-effect-based flow sensors. In some cases, the flow sensor may comprise an electromagnetic flow meter that measures flow rate based on Faraday's law of electromagnetic induction. In some cases, the flow sensor may comprise a Doppler-based ultrasonic flow meter that measures flow velocity by detecting frequency shifts in reflected ultrasonic signals. In some cases, the flow sensor may comprise a vortex flow meter that measures flow rate based on the frequency of vortices shed by a bluff body placed in the flow stream. In some cases, the flow sensor may comprise a thermal mass flow meter that measures flow rate based on heat transfer characteristics of the flowing fluid. In some cases, the flow sensor may comprise a differential pressure flow meter such as an orifice plate, venturi tube, or flow nozzle that measures flow rate based on pressure drop across a flow restriction.

[0123] Various pressure sensing technologies may be employed as alternatives to piezoresistive pressure transducers. In some cases, the pressure sensor may comprise a capacitive pressure sensor that measures pressure based on changes in capacitance between deflecting diaphragm electrodes. In some cases, the pressure sensor may comprise a strain gauge pressure sensor that measures pressure based on resistance changes in strain-sensitive elements bonded to a pressure-sensing diaphragm. In some cases, the pressure sensor may comprise a piezoelectric pressure sensor that generates electrical charge in response to applied pressure. In some cases, the pressure sensor may comprise a resonant pressure sensor that

measures pressure based on changes in resonant frequency of a vibrating element. In some cases, the pressure sensor may comprise an optical pressure sensor that measures pressure based on changes in optical properties of a sensing element.

[0124] Alternative Microcontroller Platforms

- 5 **[0125]** In some cases, the microcontroller may comprise platforms other than the ESP32 microcontroller. In some cases, the microcontroller may comprise an Arduino-based platform such as an Arduino Mega, Arduino Due, or Arduino Nano 33 IoT for applications with different processing requirements or form factor constraints. In some cases, the microcontroller may comprise a Raspberry Pi single-board computer for applications requiring greater computational resources for machine learning inference or data logging. In some cases, the microcontroller may comprise an STM32 microcontroller for applications requiring low power consumption or industrial-grade reliability. In some cases, the microcontroller may comprise a Nordic Semiconductor nRF series microcontroller for applications requiring Bluetooth Low Energy connectivity. In some cases, the microcontroller may comprise a Texas Instruments MSP430 or MSP432 microcontroller for ultra-low-power applications in battery-operated deployments.

[0126] Alternative Communication Protocols

- 20 **[0127]** In some cases, communication protocols other than MQTT may be employed for data transmission. In some cases, the system may employ HTTP or HTTPS protocols for RESTful API-based communication with cloud services. In some cases, the system may employ CoAP (Constrained Application Protocol) for lightweight machine-to-machine communication in resource-constrained environments. In some cases, the system may employ AMQP (Advanced Message Queuing Protocol) for enterprise-grade message queuing applications. In some cases, the system may employ WebSocket protocols for bidirectional real-time communication between the monitoring system and remote interfaces.

- 25 **[0128]** In some cases, wireless communication technologies other than Wi-Fi may be employed. In some cases, the system may employ LoRaWAN (Long Range Wide Area Network) for long-range, low-power communication in distributed drainage monitoring networks. In some cases, the system may employ NB-IoT

(Narrowband Internet of Things) or LTE-M cellular connectivity for wide-area coverage without dependence on local Wi-Fi infrastructure. In some cases, the system may employ Zigbee or Z-Wave mesh networking protocols for localized sensor networks with multiple monitoring nodes. In some cases, the system may
5 employ Sigfox for ultra-narrowband communication in applications requiring minimal data transmission.

[0129] Scalability for Different Drainage Infrastructure Configurations

[0130] The multi-sensor fusion system may be scaled and configured for deployment in various drainage infrastructure configurations. In some cases, the
10 system may be deployed in municipal drainage systems comprising networks of storm drains, culverts, and underground conduits serving urban areas. Municipal deployments may comprise multiple monitoring nodes distributed across the drainage network, with each monitoring node communicating with a centralized monitoring platform for network-wide surveillance and coordination.

[0131] In some cases, the system may be deployed in industrial facilities comprising process drainage systems, wastewater treatment facilities, or manufacturing plant drainage infrastructure. Industrial deployments may incorporate additional sensors for monitoring chemical parameters, temperature, or conductivity depending on the characteristics of industrial effluents. In some cases,
20 industrial deployments may integrate with existing supervisory control and data acquisition (SCADA) systems for unified facility monitoring.

[0132] In some cases, the system may be deployed in residential complexes comprising apartment buildings, housing developments, or gated communities. Residential deployments may employ simplified notification interfaces and reduced
25 sensor configurations appropriate for smaller-scale drainage infrastructure. In some cases, residential deployments may integrate with building management systems or smart home platforms for unified property monitoring.

[0133] In some cases, the system may be deployed in transportation infrastructure comprising highway drainage systems, airport runway drainage, or
30 railway track drainage. Transportation infrastructure deployments may incorporate

ruggedized sensor housings and communication systems suitable for harsh environmental conditions and vibration exposure.

[0134] In some cases, the system may employ standardized IoT data formats and ontologies for interoperability with other IoT devices and systems. In some cases, the system may support OPC UA (Open Platform Communications Unified Architecture) for integration with industrial automation systems. In some cases, the system may support Modbus or BACnet protocols for integration with building automation and control systems.

[0135] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field.

[0136] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This 25th day of April 2026

25



(Kuldeep Singh)

Authorized Agent for the Applicant,

Indian Patent Agent Regn No. IN/PA-4358

WE CLAIM:

1. A method (100) for urban drain blockage detection using multi-sensor fusion, comprising:

initializing (102) a microcontroller as a central processing and control unit
5 for real-time monitoring with wireless connectivity capabilities;

acquiring (104) real-time sensor readings from a plurality of sensors placed inside a drainage pipe, the plurality of sensors comprising:

an ultrasonic sensor configured to measure water depth by
calculating a distance between the ultrasonic sensor and a water surface in
10 the drainage pipe;

a flow sensor configured to measure a water flow rate within the drainage pipe; and

a pressure sensor configured to measure pressure changes inside the drainage pipe indicative of obstruction-induced resistance;

15 processing (106) the acquired sensor readings using threshold comparisons;
classifying (108) a system state based on the acquired sensor readings to determine one of a plurality of conditions, wherein the classifying comprises:

classifying the system state as a Critical Flood condition when the water depth exceeds a flood threshold;

20 classifying the system state as a Blockage Confirmed condition when the water flow rate decreases below a flow rate threshold AND the pressure increases above a pressure threshold simultaneously;

classifying the system state as a Warning condition when the water flow rate decreases below the flow rate threshold OR the pressure increases
25 above the pressure threshold; and

classifying the system state as a Normal condition when neither the water flow rate decreases below the flow rate threshold nor the pressure increases above the pressure threshold; and

notifying (110) a user based on the determined condition.

30 2. The method of claim 1, wherein the pressure threshold is 0.30 V and the flow rate threshold is 5.5 L/min.

3. The method of claim 1, further comprising:
sending (314) the classification results to a machine learning module for verification; and
forwarding (316) a verified result from the machine learning module to an
5 alert system.
4. The method of claim 3, wherein the machine learning module comprises a Random Forest model trained on historical drainage data.
5. The method of claim 1, wherein the notifying (110) comprises:
sending (412) a mobile notification using wireless connectivity capabilities
10 of the microcontroller for Critical and Warning conditions;
activating (414) local LED lights and a buzzer for visual and audible alerts;
and
transmitting (416) alert data to cloud services through a messaging protocol.
6. The method of claim 1, further comprising:
15 determining (208) whether the pressure exceeds the pressure threshold;
when the pressure exceeds the pressure threshold, flagging (212) the sensor readings as anomalous and forwarding data to decision-making logic; and
when the pressure does not exceed the pressure threshold, determining (210) whether the water flow rate is less than the flow rate threshold.
- 20 7. The method of claim 1, further comprising returning (112) to continuous monitoring by looping back to acquiring real-time sensor readings from the plurality of sensors.
8. The method of claim 1, wherein the ultrasonic sensor measures water depth by measuring a reflection time of ultrasonic waves emitted from the
25 ultrasonic sensor toward the water surface.
9. The method of claim 1, wherein the flow sensor comprises a Hall-effect-based flow sensor that detects rotation of an internal turbine caused by water movement within the drainage pipe.

10. The method of claim 1, wherein the pressure sensor comprises a differential pressure sensor configured to detect pressure build-up caused by pipe blockage.

Dated This 25th day of April 2026

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A handwritten signature in dark ink, appearing to read 'Kuldeep', is written over a horizontal line.

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ABSTRACT

MULTI-SENSOR FUSION SYSTEM FOR URBAN DRAIN BLOCKAGE DETECTION

A method (100) for urban drain blockage detection using multi-sensor fusion
5 comprises initializing (102) a microcontroller for real-time monitoring with
wireless connectivity capabilities and acquiring (104) real-time sensor readings
from a plurality of sensors placed inside a drainage pipe. The plurality of sensors
comprises an ultrasonic sensor configured to measure water depth by calculating a
distance between the ultrasonic sensor and a water surface, a flow sensor configured
10 to measure a water flow rate, and a pressure sensor configured to measure pressure
changes indicative of obstruction-induced resistance. The method comprises
processing (106) the acquired sensor readings using threshold comparisons and
classifying (108) a system state to determine one of a plurality of conditions
including Critical Flood, Blockage Confirmed, Warning, and Normal conditions.
15 The method comprises notifying (110) a user based on the determined condition.

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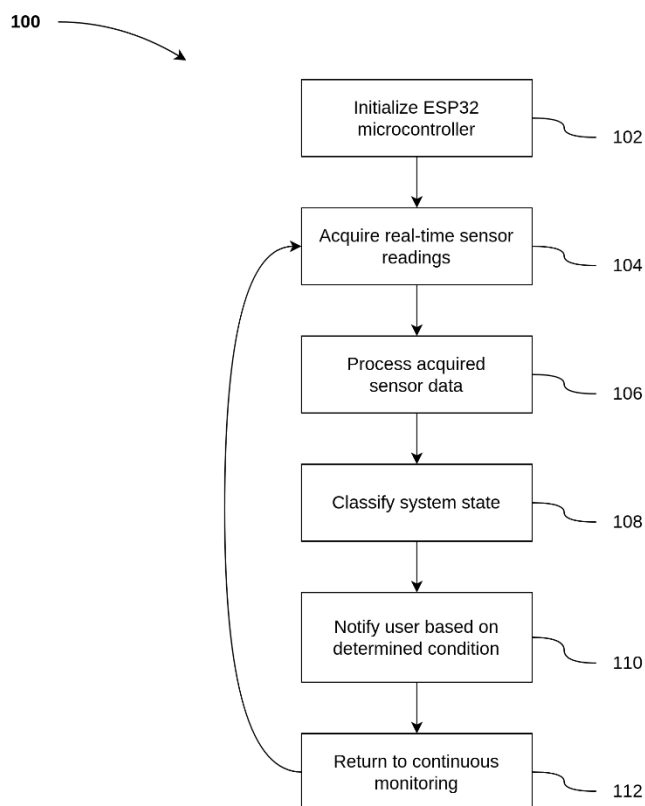


FIG. 1

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Applicant Name: VELLORE INSTITUTE OF TECHNOLOGY

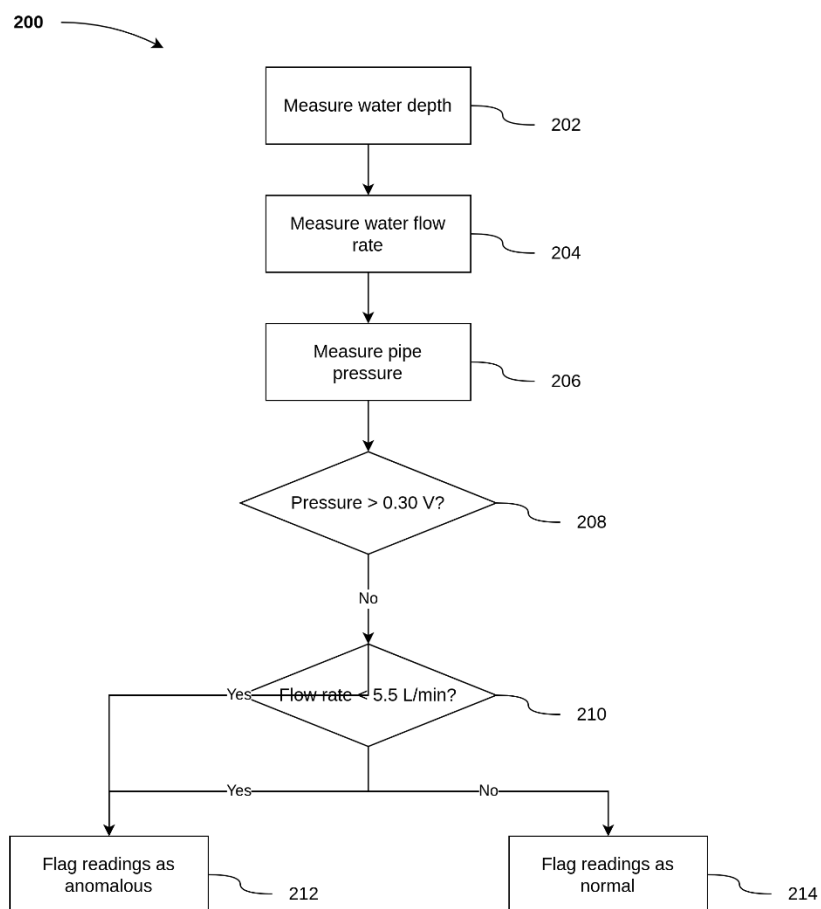


FIG. 2

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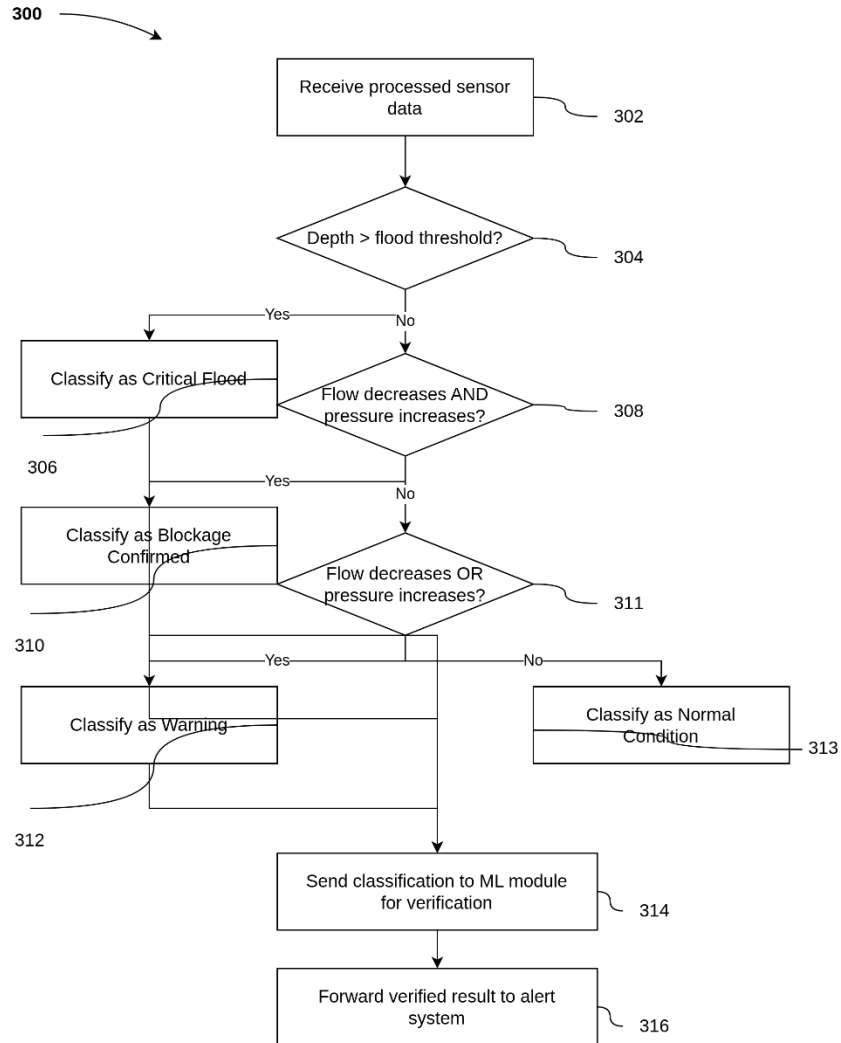


FIG. 3

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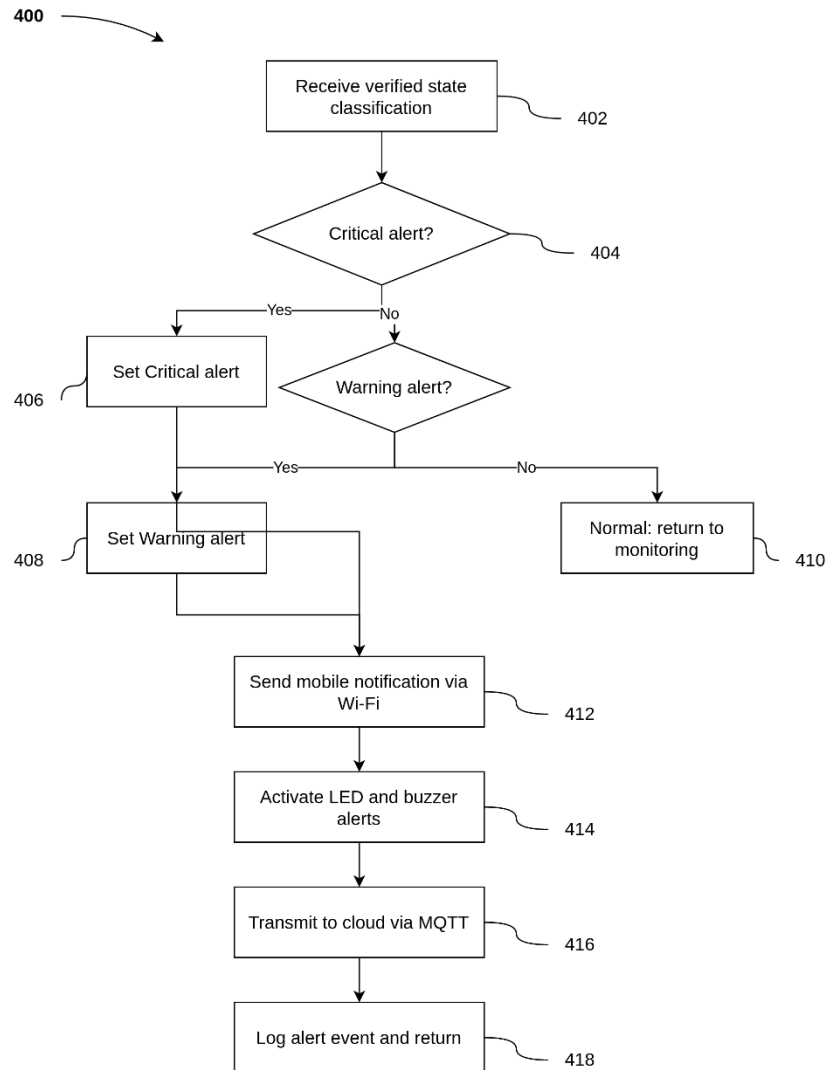


FIG. 4

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